42.510 Statistical Learning for Data Science

Linear regression

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Week 3, Lecture 2



Bordeaux wines

Bordeaux wines

- What we know –
  - Excellent weather conditions conducive for growing vines.
  - Difference in price and quality between years.
  - Aged wine is supposed to taste good.

    Younger wines not so pleasant to taste.
- Hence the importance of wine experts.
  - Robert Parker

Vintage chart 1970-2025

Robert Parker Wine Advocate

Vintage Chart 1970 - 2025

AUSTRIA	2025	2024	2023	2022	2021	2020	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010	2009	2008	2007	2006	2005	2004	2003	2002	2001	2000
AUSTRIA: NIEDERÖSTERREICH / WIEN / BURGENLAND WHITE				NT	92-95T	90-94E	93-97T	93T	92R	90R	95T	78E	91R	89T	89E	88I	89I	88R	90R	91I	87I	88I	89I	89T	88I	85C
AUSTRIA: STYRIA (STEIERMARK)				NT	NT	90-93E	92-94R	91-93R	96E	90-92R	95R	87-88E	89-94R	90-93R	NT	NT	NT	NT	NT	NT	NT	NT	NT	NT	NT	NT
AUSTRIA: NIEDERÖSTERREICH / WIEN / BURGENLAND RED				NT	NT	NT	94-96T	89-91E	93-95T	90-92R	91-94T	79-85I	90-92R	90-94R	93-95R	88-94I	93-94R	75-80R	90R	93-95R	NT	88R	88-92R	89-92R	86-90I	89-92R

FRANCE	2025	2024	2023	2022	2021	2020	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010	2009	2008	2007	2006	2005	2004	2003	2002	2001	2000
ALSACE				NT	NT	89-92T	94-96T	89-91E	93T	90E	94T	89I	86T	87R	86E	89I	82I	89R	91R	79I	87R	86E	82R	86R	91R	90R
BORDEAUX: ST JULIEN/PAUILLAC/ST ESTEPHE	96	87	94	96T	89I	96T	96T	93	91I	97T	92	95T	81C	92E	88E	98T	99E	91E	86E	87E	95T	88T	95T	88T	88R	96T
BORDEAUX: MARGAUX	94	87	93	96T	88I	95T	96T	93	90I	97T	93	93T	80C	89E	87E	95T	97E	90E	86E	88E	98T	87T	88I	88T	89E	94T
BORDEAUX: GRAVES/PESSAC LEOGNAN	94	87	94		89I	95T	96T	93	92I	97T	94	93T	81C	91E	86E	99T	98E	91E	87E	87E	96T	88T	88I	87T	88R	97T
BORDEAUX: POMEROL	94	87	94	96T	87I	95T	96T	93	91	97T	94	94T	84R	94E	88R	95T	98E	96E	86I	90T	95T	88E	84E	85E	90E	95T
BORDEAUX: ST EMILION	95	86	95	96T	88I	95T	95T	93	89I	96T	94	92T	82R	93E	87R	94T	93E	92E	86I	88E	99T	88E	90I	87E	90E	96T
BORDEAUX: BARSAC/SAUTERNES	NT	NT	97				92E	90E	91E	92E	95E	92E	92E	88E	93E	90R	97E	89R	94R	88R	96R	87R	89R	85R	93R	88E
BURGUNDY: COTE DE NUITS (RED)	NT	89	93	95	92E	97T	98E	93T	93E	97T	98T	92E	92E	95T	91E	96E	95E	88I	91E	89I	98T	83C	89R	94R	91R	87R
BURGUNDY: COTE DE BEAUNE (RED)	NT	90	93	95	91E	95T	96E	94T	92E	92E	96T	91E	89E	91E	90E	94E	95E	89I	90E	82I	96T	79C	87T	92R	88R	86R
BURGUNDY (WHITE)	NT	93	94	97	91T	97E	93T	91E	97E	87I	94E	97E	90E	92E	91E	94E	91E	94E	91I	90E	88E	91R	84R	92R	86C	88C
BURGUNDY: BEAUJOLAIS	NT	89 I	91 E	91T	90I	93E	94E	93E	94E	95I	90I	93R	86R	90R	91R	93R	97R	86R	85R	89R	95R	81C	93R	86C	75C	91R
CHAMPAGNE			92	93	86I	95T	97T	92E	87I	94T	93E	94E	95T	96T	87R	89R	92R	99T	88R	90R	88I	91R	88I	95R	88R	91R
JURA		93E	94T	91E	95E	92E	94E	93E	91I	93T	94T	93E	90I	93E	91R	94E	90R	91T	91I	90E	96R	92R	94R	93R	85R	NT
LANGUEDOC	NT	87	93	94	NT	92E	92T	91E	91T	92E	93T	87E	88E	88I	91R	94E	91R	87R	93R	90I	88R	88C	87I	NT	NT	NT
ROUSSILLON	NT	87	93	94	NT	92E	91T	90E	91T	92E	93T	88E	94E	89R	91R	94E	91R	87R	92R	90I	88R	88C	87I	NT	NT	NT

2025 2024 2023 2022 2021 2020 2019 2018 2017 2016 2015 2014 2013 2012 2011 2010 2009 2008 2007 2006 2005 2004 2003 2002 2001 2000

Vintage chart 1970-2025

RATINGS	
96-100	Extraordinary
90-95	Outstanding
80-89	Above Average to Excellent
70-79	Average
60-69	Below Average
<59	Appalling

MATURITY	
C	Caution, may be too old
E	Early maturing and accessible
NV	Vintage not declared
I	Irregular, even among the best wines
NT	Not yet sufficiently tested to rate
R	Ready to drink
T	Still tannic, youthful, or slow to mature

The Statistical Sommelier

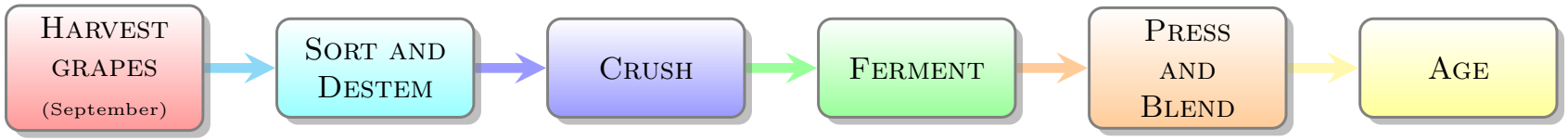
- Orley Ashenfelter, Professor of Economics, Princeton University.
- Claims that he can predict quality without tasting.
- Tool used: Linear Regression.
- Liquid asset

Why predict?

- . . . ???

- Investment: *en primeur* or *wine futures*
- tasting samples of wine within a year or two after it is made.
- The 1982 vintage wine of Chateau Latour that was sold at 250 pounds a case *en primeur* in 1983 was valued at 9000 British pounds in 2007.
- Video

The wine making process



Insights

Ashenfelter focused on the vintage as a predictor for the quality of wine by averaging auction prices across chateaus.

Observations

- Older the wine is, the greater is the value.
- There is still significant variation in average prices that remains unexplained.

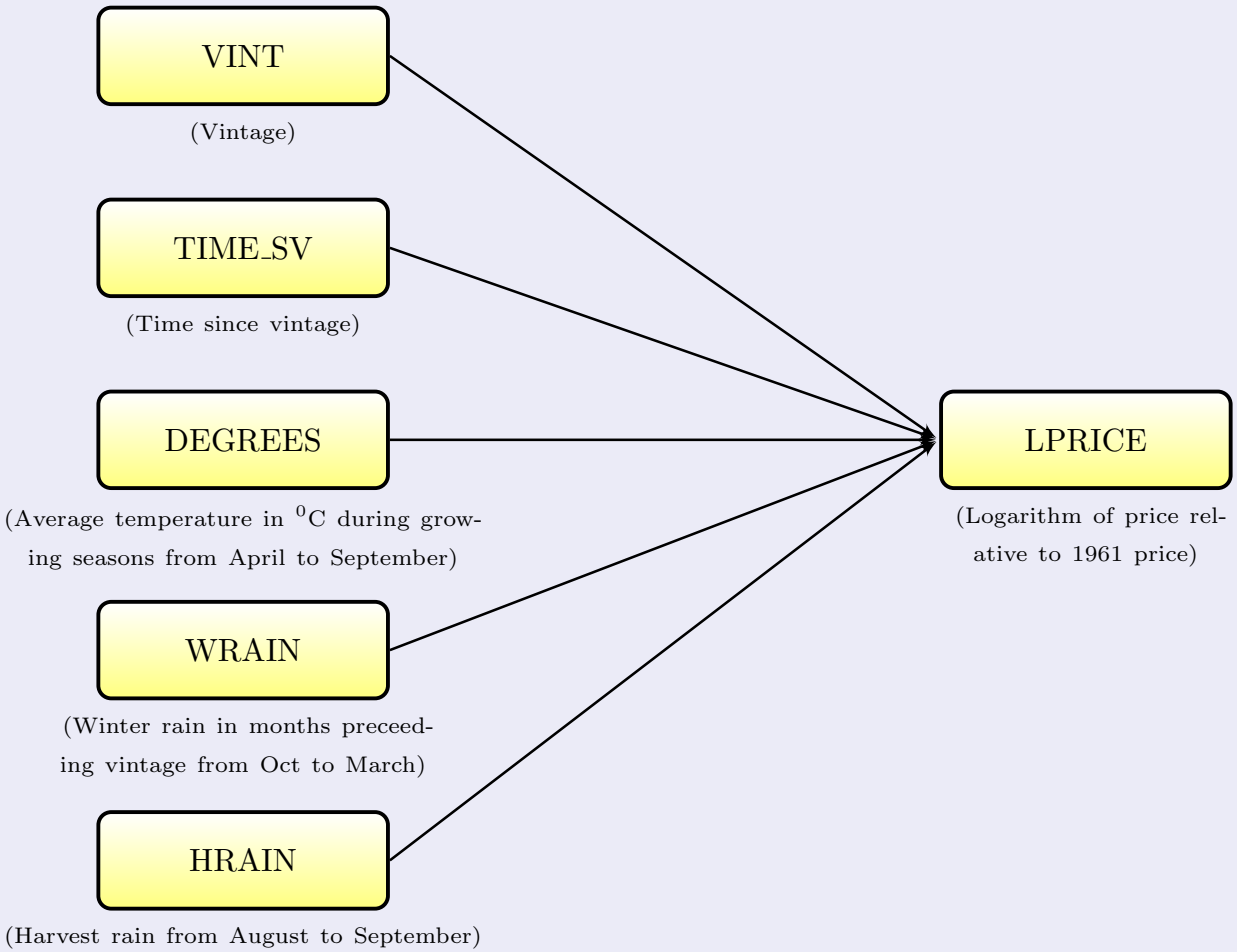
Possible predictors

- Vineyard (chateau - location where wine is made)
- Vintage (year - time when wine is made)

Wine making in Bordeaux

Datasets and variables

Years: 1952 - 1989



- Variables used from data in Liquid asset

Exercise

Suppose price is measured on the log scale and the fitted model is

$$\widehat{\log(\mathit{Price})} = a + 0.08\,\mathit{Age} + 0.03\,\mathit{Temp} - 0.002\,\mathit{HarvestRain}.$$

1. Holding other variables fixed, interpret the coefficient of `Temp`.
2. Compare two vintages which are otherwise identical except for harvest rainfall which differs by 50 mm.
3. Identify one important source of uncertainty that is absent from the equation.

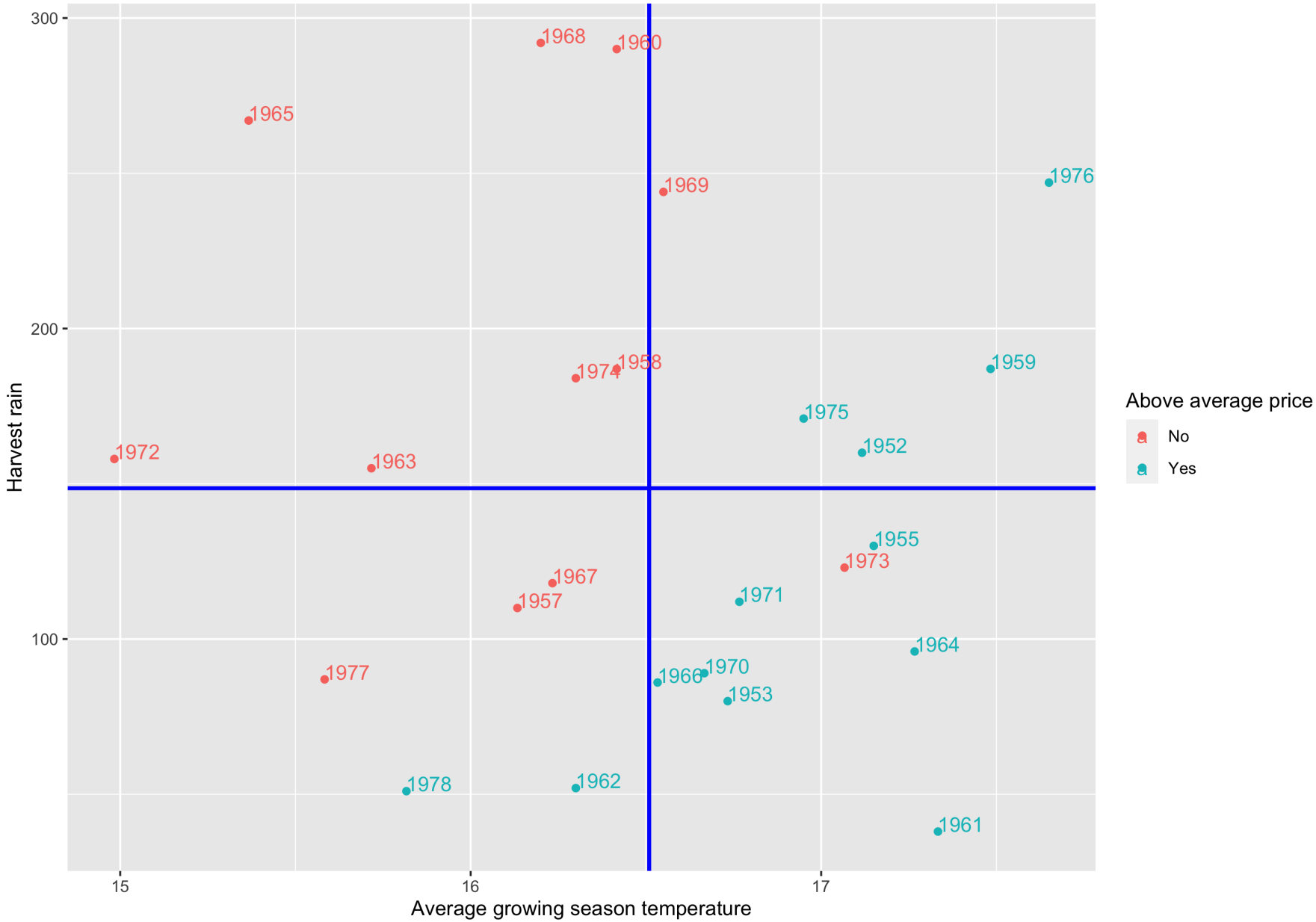
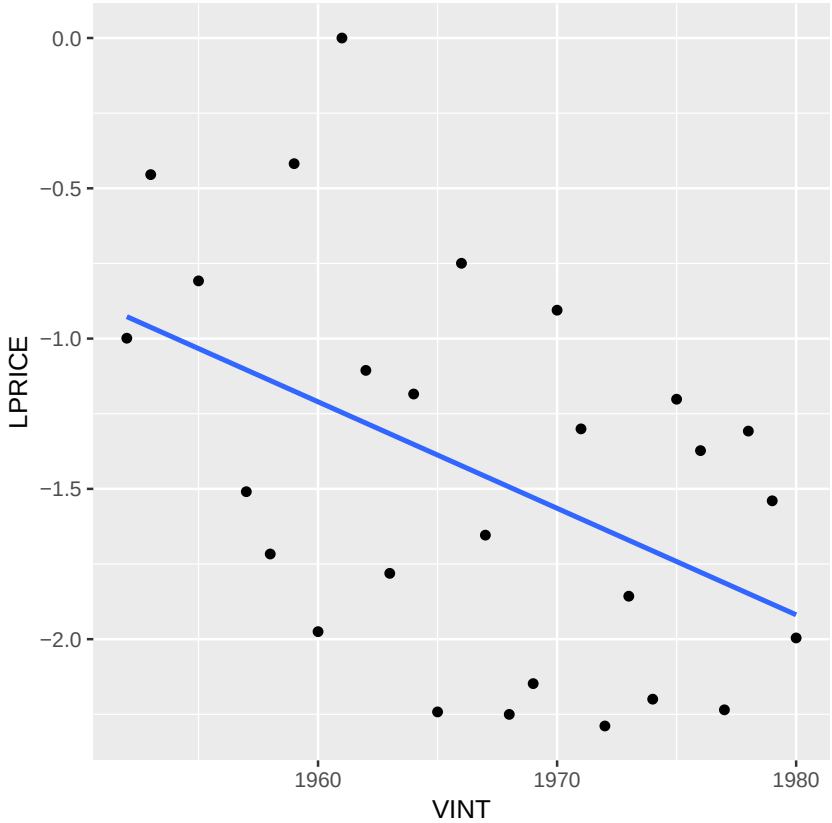


Figure 1: Harvest Rain vs Temperature

Linear regression



Problem setup

- n = Number of observations
- p = Number of predictor variables
- y = Dependent variable in $\mathbb{R}$
- $x_1, \ldots, x_p$ = Independent variables (predictors)

- Want to predict y with a function $f(\mathbf{x})$.
- Let $f(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p$.
- Data: $(y_i, x_{i1}, \ldots, x_{ip})$, $i = 1, \ldots, n$.
- Method of least squares:

$$\min_{\beta_0,\ldots,\beta_p} \frac{1}{n} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \ldots - \beta_p x_{ip})^2.$$

- This is Empirical Risk minimization (ERM).
- Minimize loss L (in expectation) where

$$\mathbb{E}L(Y, f(\mathbf{X})) = \mathbb{E}(Y - f(\mathbf{X}))^2$$

  with some assumptions on $\mathbb{P}(\mathbf{X}, Y)$.
- Structural assumption: $f(\mathbf{x}) = \mathbf{x}^\top \beta$.

Linear regression: Notation and set-up

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbb{X} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}.$$

- We want to minimize $Q(\boldsymbol{\beta})$ w.r.t $\boldsymbol{\beta}$ where

$$Q(\boldsymbol{\beta}) = (\mathbf{y} - \mathbb{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbb{X}\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbb{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbb{X}^\top \mathbb{X}\boldsymbol{\beta}$$

- Taking derivatives and equating to zero we have obtain:

$$\widehat{\boldsymbol{\beta}} = (\mathbb{X}^\top \mathbb{X})^{-1} \mathbb{X}^\top \mathbf{y}.$$

- *Fitted values:* $\widehat{y}_i = \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}$.
- *Residuals:* $e_i = y_i - \widehat{y}_i$.
- Note that $Q(\boldsymbol{\beta})$ is quadratic in $\boldsymbol{\beta}$. This is a convex problem, so every local minimum is also a global minimum.

Linear regression

- No statistical assumptions until this. But we would like to compute standard error (variability in the estimates).
- Assume: $\mathbf{Y} = \mathbb{X}\beta + \boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$.
- Therefore $\text{Var}(\mathbf{Y}) = \text{Var}(\mathbb{X}\beta + \boldsymbol{\epsilon}) = \text{Var}(\boldsymbol{\epsilon}) = \sigma^2\mathbf{I}$.
- Since for any $k \times n$ matrix $\mathbf{A}$, we have $\text{Var}(\mathbf{AY}) = \sigma^2\mathbf{A}\mathbf{A}^\top$, we get

$$\text{Var}(\widehat{\beta}) = \sigma^2(\mathbb{X}^\top\mathbb{X})^{-1}$$

- Hence we have standard error: $\mathbf{se}(\widehat{\beta}_j) = \sigma\sqrt{((\mathbb{X}^\top\mathbb{X})^{-1})_{jj}}$.
- σ^2 is unknown but we can obtain an unbiased estimate of σ^2 from the data:

$$\hat{\sigma}^2 = \frac{1}{n-p-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1}{n-p-1}\sum_{i=1}^{n} e_i^2 = \frac{Q(\widehat{\beta})}{n-p-1}$$

where $Q(\widehat{\beta})$ is residual sum of squares (RSS) or sum of squared errors (SSE).

Linear regression: important questions

- Is X_j useful in predicting the response Y?
- How well does the model fit the data?

Linear regression: Is the j-th variable significant?

- A natural question here: is the jth variable useful in the model?
- We can conduct a one-sample t-test to ascertain this.
- We intend to test:

$$\mathbf{H}_0 : \beta_j = 0 \qquad \text{vs.} \qquad \mathbf{H}_1 : \beta_j \neq 0$$

- Since $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$, we have obtained $\widehat{\beta}_j \sim \mathcal{N}(\beta_j, \sigma^2((\mathbb{X}^\top\mathbb{X})^{-1})_{jj})$.
- Under $\mathbf{H}_0$, we have $\beta_j = 0$, so $\widehat{\beta}_j \sim \mathcal{N}(0, \sigma^2((\mathbb{X}^\top\mathbb{X})^{-1})_{jj})$
- Therefore under $\mathbf{H}_0$ we have the test statistic

$$t_j = \frac{\widehat{\beta}_j}{\hat{\sigma}\sqrt{((\mathbb{X}^\top\mathbb{X})^{-1})_{jj}}} \sim t_{n-p-1},$$

  a t-distribution with $(n - p - 1)$ degrees of freedom.
- We reject $\mathbf{H}_0$ for very high (absolute) values of the test statistic $|t_j|$ or very small p-values.

Linear regression: Quality of fit

- Note that for the regression model the following quantities are important:
- Sum of squared errors (SSE) or Residual sum of squares (RSS):

$$\text{SSE} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

- Sum of squares due to regression (SSR):

$$\text{SSR} = \sum_{i=1}^{n} (\hat{y}_i - \overline{y})^2$$

- Total sum of squares (SST):

$$\text{SST} = \sum_{i=1}^{n} (y_i - \overline{y})^2$$

- We also have the following:

$$\text{SST} = \text{SSR} + \text{SSE}$$

Linear regression: Quality of fit

- Coefficient of Determination:

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}.$$

- Check that $0 \leq R^2 \leq 1$.
- $R^2 = 0$: the linear regression model has no explanatory power.
- $R^2 = 1$: the linear regression model is a perfect predictor.
- If $Y = \beta_0 + \beta_1 X_1$ then $R^2 = (\text{corr}(X_1, Y))^2$.
- As the number of variables $p \uparrow$, we have $R^2 \uparrow$ as well.

- *Alternative measure*: Adjusted-R^2:

$$\text{Adjusted-}R^2 = 1 - (1 - R^2)\frac{n-1}{n-p-1}$$

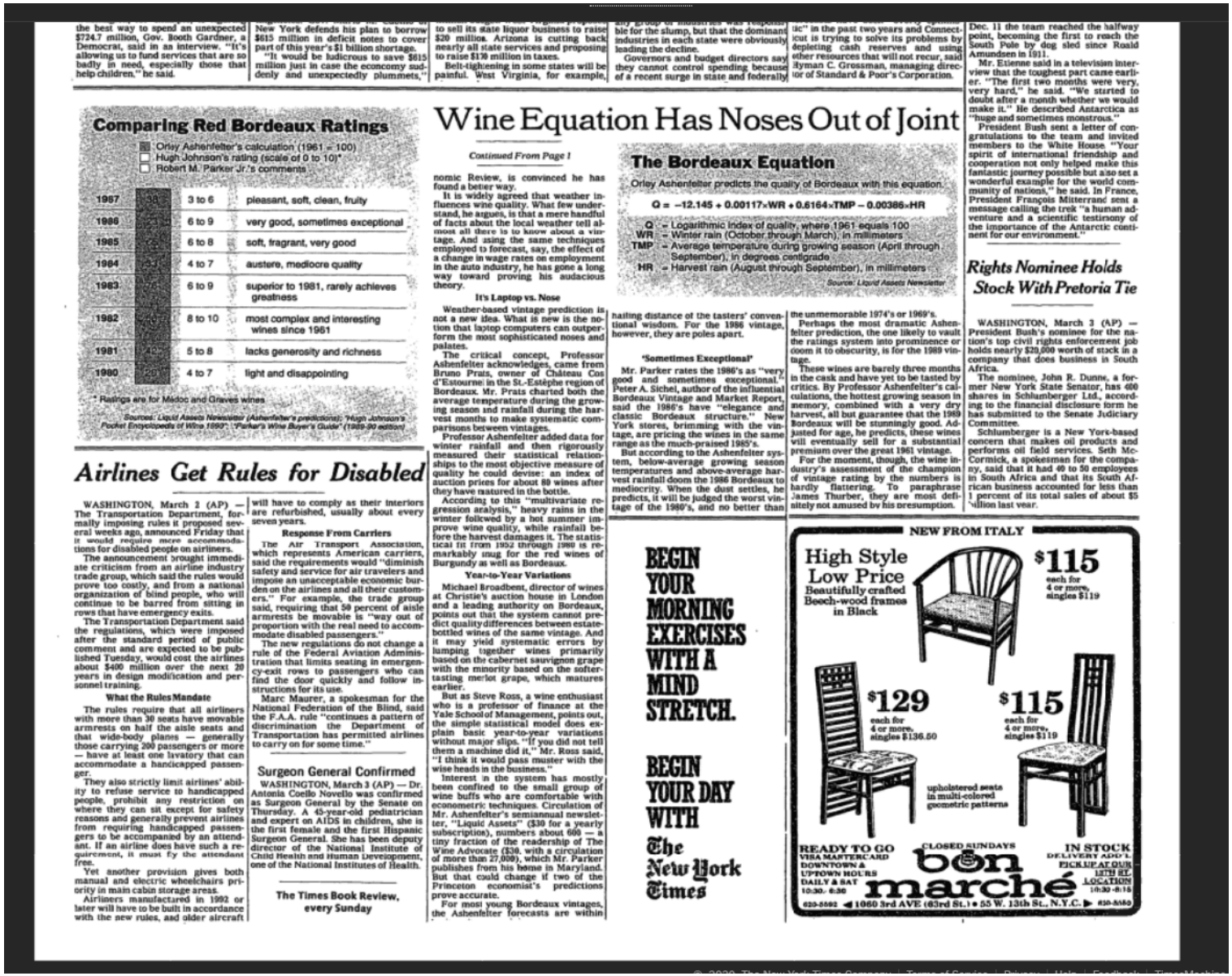
It penalizes for adding more variables.

Linear regression: other important questions

- Is the model significant?
- Are all variables important?
- How good is our prediction?
- We will address these, and more such pertinent questions in the next class.
- First we do some analysis in R.

R workflow: a few basic commands

```
wine <- read.csv("wine.csv")
fit <- lm(
  LPRICE ~ VINT+WRAIN + HRAIN + DEGREES,
  data = winetrain
)
summary(fit)
plot(fit)
predict(fit, newdata = winetest, interval = "prediction")
```



the best way to spend an unexpected $724.7 million, Gov. Booth Gardner, a Democrat, said in an interview. "It's allowing us to fund services that are so badly in need, especially those that help children," he said.

New York defends his plan to borrow $615 million in deficit notes to cover part of this year's $1 billion shortage.

"It would be ludicrous to save $615 million just in case the economy suddenly and unexpectedly plummets,"

to sell its state liquor business to raise $20 million. Arizona is cutting back nearly all state services and proposing to raise $130 million in taxes.

Belt-tightening in some states will be painful. West Virginia, for example,

ble for the slump, but that the dominant industries in each state were obviously leading the decline.

Governors and budget directors say they cannot control spending because of a recent surge in state and federally

ic" in the past two years and Connecticut is trying to solve its problems by depleting cash reserves and using other resources that will not recur, said Hyman C. Grossman, managing director of Standard & Poor's Corporation.

Dec. 11 the team reached the halfway point, becoming the first to reach the South Pole by dog sled since Roald Amundsen in 1911.

Mr. Etienne said in a television interview that the toughest part came earlier. "The first two months were very, very hard," he said. "We started to doubt after a month whether we would make it." He described Antarctica as "huge and sometimes monstrous."

President Bush sent a letter of congratulations to the team and invited members to the White House. "Your spirit of international friendship and cooperation not only helped make this fantastic journey possible but also set a wonderful example for the world community of nations," he said. In France, President François Mitterrand sent a message calling the trek "a human adventure and a scientific testimony of the importance of the Antarctic continent for our environment."

Comparing Red Bordeaux Ratings

Orley Ashenfelter's calculation (1961 = 100)
Hugh Johnson's rating (scale of 0 to 10)*
Robert M. Parker Jr.'s comments

1987	93	3 to 6	pleasant, soft, clean, fruity
1986	96	6 to 9	very good, sometimes exceptional
1985	92	6 to 8	soft, fragrant, very good
1984	93	4 to 7	austere, mediocre quality
1983	96	6 to 9	superior to 1981, rarely achieves greatness
1982	100	8 to 10	most complex and interesting wines since 1961
1981	92	5 to 8	lacks generosity and richness
1980	90	4 to 7	light and disappointing

* Ratings are for Médoc and Graves wines

Sources: Liquid Assets Newsletter (Ashenfelter's predictions); "Hugh Johnson's Pocket Encyclopedia of Wine 1990"; "Parker's Wine Buyer's Guide" (1989-90 edition)

Wine Equation Has Noses Out of Joint

Continued From Page 1

nomic Review, is convinced he has found a better way.

It is widely agreed that weather influences wine quality. What few understand, he argues, is that a mere handful of facts about the local weather tell almost all there is to know about a vintage. And using the same techniques employed to forecast, say, the effect of a change in wage rates on employment in the auto industry, he has gone a long way toward proving his audacious theory.

It's Laptop vs. Nose

Weather-based vintage prediction is not a new idea. What is new is the notion that laptop computers can outperform the most sophisticated noses and palates.

The critical concept, Professor Ashenfelter acknowledges, came from Bruno Prats, owner of Château Cos d'Estournel in the St.-Estèphe region of Bordeaux. Mr. Prats charted both the average temperature during the growing season and rainfall during the harvest months to make systematic comparisons between vintages.

Professor Ashenfelter added data for winter rainfall and then rigorously measured their statistical relationships to the most objective measure of quality he could devise: an index of auction prices for about 80 wines after they have matured in the bottle.

According to this "multivariate regression analysis," heavy rains in the winter followed by a hot summer improve wine quality, while rainfall before the harvest damages it. The statistical fit from 1952 through 1980 is remarkably snug for the red wines of Burgundy as well as Bordeaux.

Year-to-Year Variations

Michael Broadbent, director of wines at Christie's auction house in London and a leading authority on Bordeaux, points out that the system cannot predict quality differences between estate-bottled wines of the same vintage. And it may yield systematic errors by lumping together wines primarily based on the cabernet sauvignon grape with the minority based on the softer-tasting merlot grape, which matures earlier.

But as Steve Ross, a wine enthusiast who is a professor of finance at the Yale School of Management, points out, the simple statistical model does explain basic year-to-year variations without major slips. "If you did not tell them a machine did it," Mr. Ross said, "I think it would pass muster with the wise heads in the business."

Interest in the system has mostly been confined to the small group of wine buffs who are comfortable with econometric techniques. Circulation of Mr. Ashenfelter's semiannual newsletter, "Liquid Assets" ($30 for a yearly subscription), numbers about 600 — a tiny fraction of the readership of The Wine Advocate ($30, with a circulation of more than 27,000), which Mr. Parker publishes from his home in Maryland. But that could change if two of the Princeton economist's predictions prove accurate.

For most young Bordeaux vintages, the Ashenfelter forecasts are within hailing distance of the tasters' conventional wisdom. For the 1986 vintage, however, they are poles apart.

'Sometimes Exceptional'

Mr. Parker rates the 1986's as "very good and sometimes exceptional." Peter A. Sichel, author of the influential Bordeaux Vintage and Market Report, said the 1986's have "elegance and classic Bordeaux structure." New York stores, brimming with the vintage, are pricing the wines in the same range as the much-praised 1985's.

But according to the Ashenfelter system, below-average growing season temperatures and above-average harvest rainfall doom the 1986 Bordeaux to mediocrity. When the dust settles, he predicts, it will be judged the worst vintage of the 1980's, and no better than the unmemorable 1974's or 1969's.

Perhaps the most dramatic Ashenfelter prediction, the one likely to vault the ratings system into prominence or doom it to obscurity, is for the 1989 vintage.

These wines are barely three months in the cask and have yet to be tasted by critics. By Professor Ashenfelter's calculations, the hottest growing season in memory, combined with a very dry harvest, all but guarantee that the 1989 Bordeaux will be stunningly good. Adjusted for age, he predicts, these wines will eventually sell for a substantial premium over the great 1961 vintage.

For the moment, though, the wine industry's assessment of the champion of vintage rating by the numbers is hardly flattering. To paraphrase James Thurber, they are most definitely not amused by his presumption.

The Bordeaux Equation

Orley Ashenfelter predicts the quality of Bordeaux with this equation.

$$Q = -12.145 + 0.00117 \times WR + 0.6164 \times TMP - 0.00386 \times HR$$

Q – Logarithmic index of quality, where 1961 equals 100
WR – Winter rain (October through March), in millimeters
TMP – Average temperature during growing season (April through September), in degrees centigrade
HR – Harvest rain (August through September), in millimeters

Source: Liquid Assets Newsletter

Airlines Get Rules for Disabled

WASHINGTON, March 2 (AP) — The Transportation Department, formally imposing rules it proposed several weeks ago, announced Friday that it would require more accommodations for disabled people on airliners.

The announcement brought immediate criticism from an airline industry trade group, which said the rules would prove too costly, and from a national organization of blind people, who will continue to be barred from sitting in rows that have emergency exits.

The Transportation Department said the regulations, which were imposed after the standard period of public comment and are expected to be published Tuesday, would cost the airlines about $400 million over the next 20 years in design modification and personnel training.

What the Rules Mandate

The rules require that all airliners with more than 30 seats have movable armrests on half the aisle seats and that wide-body planes — generally those carrying 200 passengers or more — have at least one lavatory that can accommodate a handicapped passenger.

They also strictly limit airlines' ability to refuse service to handicapped people, prohibit any restriction on where they can sit except for safety reasons and generally prevent airlines from requiring handicapped passengers to be accompanied by an attendant. If an airline does have such a requirement, it must fly the attendant free.

Yet another provision gives both manual and electric wheelchairs priority in main cabin storage areas.

Airliners manufactured in 1992 or later will have to be built in accordance with the new rules, and older aircraft will have to comply as their interiors are refurbished, usually about every seven years.

Response From Carriers

The Air Transport Association, which represents American carriers, said the requirements would "diminish safety and service for air travelers and impose an unacceptable economic burden on the airlines and all their customers." For example, the trade group said, requiring that 50 percent of aisle armrests be movable is "way out of proportion with the real need to accommodate disabled passengers."

The new regulations do not change a rule of the Federal Aviation Administration that limits seating in emergency-exit rows to passengers who can find the door quickly and follow instructions for its use.

Marc Maurer, a spokesman for the National Federation of the Blind, said the F.A.A. rule "continues a pattern of discrimination the Department of Transportation has permitted airlines to carry on for some time."

Surgeon General Confirmed

WASHINGTON, March 3 (AP) — Dr. Antonia Coello Novello was confirmed as Surgeon General by the Senate on Thursday. A 45-year-old pediatrician and expert on AIDS in children, she is the first female and the first Hispanic Surgeon General. She has been deputy director of the National Institute of Child Health and Human Development, one of the National Institutes of Health.

The Times Book Review, every Sunday

Rights Nominee Holds Stock With Pretoria Tie

WASHINGTON, March 3 (AP) — President Bush's nominee for the nation's top civil rights enforcement job holds nearly $20,000 worth of stock in a company that does business in South Africa.

The nominee, John R. Dunne, a former New York State Senator, has 400 shares in Schlumberger Ltd., according to the financial disclosure form he has submitted to the Senate Judiciary Committee.

Schlumberger is a New York-based concern that makes oil products and performs oil field services. Seth McCormick, a spokesman for the company, said that it had 49 to 50 employees in South Africa and that its South African business accounted for less than 1 percent of its total sales of about $5 billion last year.

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Link

The Bordeaux equation and wine critics

- In 1991, Ashenfelter predicted 1989 and 1990 to be exceptional years. Wine critics did not agree at that time; but there is almost unanimous agreement now that they are the best vintages of the last 50 years.
- Ashenfelter's model and Parker's expert opinion both agree that the 2000 and 2003 vintage would be very high quality wine.
- Ashenfelter predicted the 1989 vintage to be excellent and 1990 even better. At first Robert Parker predicted this to be similar to the 1985 vintage but then later said it was the vintage of the century.
- All of this was done without tasting a single drop of wine.

How can it make my life better?

Top 10 Bordeaux Vintages

The top 10 Bordeaux vintages as predicted by the wine weather quality index is shown below.

Top 10 Bordeaux Vintages by Year

Based on a Wine Quality Weather Index

Year	Index Value
2003	1.128
2020	1.070
2018	1.061
2011	0.996
2006	0.856
2014	0.746
1961	0.721
2016	0.682
2019	0.602
1982	0.483

Top 10 Bordeaux Vintages based on the Wine Quality Index

The Ashenfelter wine quality index creates waves of controversy when it was first introduced. This post reevaluated the index while correcting for a few issues in an effort to improve it. Once reconstituted, we can calculate the top 10 Bordeaux vintages.

Courtesy: World Wine Weather

Points to ponder

Suppose your model has $R^2 = 0.828$ (as observed). A plot of the residuals shows a strong curve.

1. Is the model adequate? Why or why not?
2. What is a plausible next step?
3. Which output would you report for a future vintage: a confidence interval or a prediction interval?

References

- Ashenfelter (1990), *Predicting the Quality and Prices of Bordeaux Wine*.
- Hastie et al. (2001), *The Elements of Statistical Learning*, Chapter 3.
- James et al. (2023), *Introduction to Statistical Learning*, Chapter 3.